

Javascript - window object

- ⊛ Window object ବନ୍ଧିବା ସମୟରେ ବାବୁ; browser ଥିବା open window ଥିବୁ.
- ⊛ If a document contain frames (<iframe> tags), the browser creates one window object for the HTML document, and one additional window object for each frame.

addEventListener()

- ⊛ this method attached an event handler to a window.

ଉଦା:- <p> Click anywhere in the window to display "Hello world." </p>
<p id="demo"></p>

<script>

window.addEventListener("click", myfunction);

function myfunction () {

document.getElementById("demo").innerHTML =
"Hello World.";

}

</script>

★ ଚେନି ଏ window ଥିବାକୁ କ୍ଲିକ୍ କଲେ ଏହାକୁ click କଲେ ହିଁ;
"Hello World." ଡିସପ୍ଲେ ହେବୁ.

removeEventListener()

- ⊛ this method removes an event handler from a document.

- ⊛ Syntax;

document.removeEventListener(event, function, capture)

★ event - Required. The name of the event to remove.

Do not use "on" prefix. use "click" instead of "onclick".

★ function - Required. The function to remove.

★ capture - Optional. (Default ⇒ false)

true - Remove the handler from capturing

false - Remove the handler from bubbling

If the event handler was attached two times, one with capturing and one with bubbling, each must be removed separately.

2201- <P> A mousemove event handler displays a random number every time you move your mouse in this document. </P>

<P> Click "Remove" to remove the event handler. </P>

<button onclick = "removeHandler()"> Remove </button>

<P id = "demo"> </P>

<script>

~~document.getElementById~~

document.addEventListener("mousemove", myFunction);

function myFunction() {

document.getElementById("demo").innerHTML = Math.random();

}

function removeHandler() {

document.removeEventListener("mousemove", myFunction);

}

</script>

alert()

⊛ this method displays an alert box with a message and a OK button.

2201- <script>

alert("Hello");

</script>

atob() , btoa()

⊛ atob() method decodes a ~~string~~ base-64 encoded string.

⊛ btoa() method encodes a ~~base~~ string in base-64

2201- <P id = "demo"> </P>

<script>

let text = "Hello World!";

let encoded = window.btoa(text);

let decoded = window.atob(encoded);

document.getElementById("demo").innerHTML =

"Encoded: " + encoded + "
" + "Decoded: " + decoded;

</script>

blur()

- ⊛ this method removes focus from a window.

```
eg:- <P> Click to open a new window, and blur it (remove focus) </P>  
<button onclick = "myFunction()" > Click Me! </button>  
<script>  
  function myFunction() {  
    var myWindow = window.open("", "", "width=200,height=100");  
    myWindow.blur();  
  }  
</script>
```

clearInterval()

- ⊛ this method clears a timer set with the setInterval() method.

Note:

- ⊛ To clear an interval, use the id returned from setInterval():
myInterval = setInterval(function, milliseconds);

- ⊛ Then you can stop the execution by calling clearInterval():
clearInterval(myInterval);

```
eg:- <P id = "demo"> </P>  
<button onclick = "myStop()" > Stop </button>  
<script>  
  const myInterval = setInterval(myTimer, 1000);  
  function myTimer() {  
    const date = new Date();  
    document.getElementById("demo").innerHTML =  
      date.toLocaleTimeString();  
  }  
  function myStop() {  
    clearInterval(myInterval);  
  }  
</script>
```

open(), close() and closed

- ⊛ open() method opens a new browser window, or a new tab, depending on the browser settings and the parameter values.

④ open() Syntax;

window.open(URL, name, specs, replace)

* URL - optional. The URL of the page to open. If no URL is specified, a new blank window/tab is opened.

* name - optional. The target attribute or the name of the window. The following values are supported;

value	Description
_blank	URL is loaded into a new window. This is default
_parent	URL is loaded into the parent frame.
_self	URL replaces the current page.
_top	URL replaces any framesets that may be loaded.

* specs - optional. A comma-separated list of items, no whitespaces.

* replace - Deprecated. Specifies whether the URL creates a new entry or replaces the current entry in the history list. The following values are supported;

- true - URL replaces the current document in the history list

- false - URL creates a new entry in the history list

```
<<!-- <button onclick = "myFunction()" > Click Me! </button>
</script>
```

```
function myFunction () {
```

```
    window.open ("https://www.w3schools.com", "_blank",
```

```
    "toolbar = yes, scrollbars = yes, resizable = yes, top = 500, left = 500,
```

```
    width = 400, height = 400");
```

```
}
```

```
</script>
```

④ close() method closes a window.

④ close() Syntax;

window.close()

```
<<!-- <button onclick = "openWin()" > Open in a new window </button>
```

```
<button onclick = "closeWin()" > Close the new window </button>
```

```
<script>
```

```
let myWindow;
```

```
function openWin() {
```

```
  myWindow = window.open("https://www.w3schools.com",  
    "_blank", "width = 500, height = 500");
```

```
}
```

```
function closeopenWin() {
```

```
  myWindow.close();
```

```
}
```

```
</script>
```

- ⊛ closed property returns 'true' if the window is closed. closed is read-only.

220:- <button onclick = "checkWin()"> Check </button>

```
<script>
```

```
let myWindow;
```

```
if (!myWindow)
```

```
function checkWin() {
```

```
  let text = "";
```

```
  if (!myWindow) {
```

```
    text = "It has never been opened!";
```

```
  }
```

```
  else {
```

```
    if (myWindow.closed) {
```

```
      text = "Closed!";
```

```
    }
```

```
    else {
```

```
      text = "Opened";
```

```
    }
```

```
  }
```

```
  alert(text);
```

```
}
```

```
</script>
```

confirm()

- ⊛ this method displays a dialog box with a message, an OK button and a Cancel button.
- ⊛ this method returns 'true' if the user clicked "OK", otherwise 'false'.

```

<<?> <button onclick = "myfunction()"> Click Me! </button>
<script>
  function myfunction () {
    confirm ("Press a button!");
  }
</script>

```

frameElement

- ⊛ this property returns the frame where the window runs.
- ⊛ this one returns 'null' if the window does not run in a frame.
- ⊛ this is a read-only property.
- ⊛ Syntax;

```

window.frameElement
or
frameElement

```

```

<<?> <p id = "demo"></p>
<script>
  let answer = "NO";
  if (window.frameElement) {
    answer = "YES";
  }
  document.getElementById ("demo").innerHTML = answer;
</script>

```

frames

- ⊛ this property returns an array with all window objects in the window.
- ⊛ this is a read-only property and those windows can be accessed by index numbers.

```

<<?> <button onclick = "myfunction()"> Click Me! </button> <br><br>
<iframe src = "https://www.w3schools.com"></iframe>
<iframe src = "https://www.w3schools.com"></iframe>
<script>
  function myfunction () {
    window.frames[0].location = "https://www.w3schools.com/jsref/";
  }
</script>

```


history

- ⊛ this object contains the URLs visited by user. (in the browser window)
- ⊛ history object uses following methods;

Method	Description
back()	Loads the previous URL in the history list
forward()	Loads the next URL in the history list
go()	Loads a specific URL from the history list
length	Returns the number of URLs in the history

Ex-1- `<p id="demo"></p>`

`<script>`

`let length = history.length;`

`document.getElementById("demo").innerHTML = length;`

`</script>`

getComputedStyle()

- ⊛ this method gets the computed CSS properties and values of an HTML element.
- ⊛ this method returns `CSSStyleDeclaration` object.
- ⊛ Syntax;

`window.getComputedStyle(element, pseudoElement)`

* element - Required. The element to ~~be~~ get the computed style for.

* pseudo element - Optional. A pseudo-element to get.

~~`<p id="demo"></p>`~~

~~`<script>`~~

~~`const element = document.getElementById(`~~

innerHeight, innerWidth

- ⊛ `innerHeight` property returns the height of a window's content area.
this is a read-only property.
- ⊛ `innerWidth` property returns the width of a window's content area.
this is a read-only property.

eg:- <P id="demo"></P>

<script>

let w = window.innerWidth;

let h = window.innerHeight;

document.getElementById("demo").innerHTML =
"Width: " + w + "
Height: " + h;

</script>

outerHeight, outerWidth

- ④ outerHeight property returns the outer height of the browser window, including all interface elements (like toolbars / scrollbars). read-only property.
- ④ outerWidth property returns the outer width of the browser window, including all interface elements (like toolbars / scrollbars). read-only property.

eg:- <P id="demo"></P>

<script>

let w = window.outerWidth;

let h = window.outerHeight;

document.getElementById("demo").innerHTML =
"Width: " + w + "
Height: " + h;

</script>

length

- ④ This property returns the number of (framed) windows in the window. Read-only property. windows can be accessed by index numbers.

eg:- <P id="demo"></P>

<iframe style="width:100%; height:100%;"></iframe>

<iframe style="width:100%; height:100%;"></iframe>

<script>

let length = window.length;

document.getElementById("demo").innerHTML = length;

</script>

localStorage

- ⊛ this object allows us to save key/value pairs in the browser.
- ⊛ Syntaxes;

window.localStorage or localStorage

- * Save data to local storage;

```
localStorage.setItem(key, value);
```

- * Read data from local storage;

```
let lastname = localStorage.getItem(key);
```

- * Remove data from local storage;

```
localStorage.removeItem(key);
```

- * Clear local storage;

```
localStorage.clear();
```

- * key - Required. The name of a key.

- * value - Required. The value of the key.

```
<<<- <P id="demo"></P>
```

```
<script>
```

```
// Set item
```

```
localStorage.setItem("lastname", "Smith");
```

```
// Retrieve
```

```
document.getElementById("demo").innerHTML =  
localStorage.getItem("lastname");
```

```
</script>
```

location

- ⊛ this object contains information about the current URL.
- ⊛ this is a property of the window object.

```
<<<- <P id="demo"></P>
```

```
<script>
```

```
let origin = url window.location.origin;
```

```
document.getElementById("demo").innerHTML = origin;
```

```
</script>
```

Ⓐ Location object properties;

property	Description
hash	Sets or returns the anchor part (#) of URL
host	Sets or returns the hostname and port number of a URL
hostname	Sets or returns the hostname of a URL
href	Sets or returns the entire URL
origin	Returns, the protocol, hostname and port number of a URL
pathname	Sets or returns the path name of a URL
port	Sets or returns the port number of a URL
protocol	Sets or returns the protocol of a URL
search	Sets or returns the querystring part of a URL

Ⓐ Location object methods;

Method	Description
assign()	Loads a new document
reload()	Reloads the current document
replace()	Replaces the current document with a new one

moveBy()

Ⓐ this method moves a window a number of pixels relative to its current coordinates.

Ⓐ Syntax;

window.moveBy(x,y)

- * x - Required. A positive or negative number. The number of pixels to move the window horizontally.
- * y - Required. A positive or negative number. The number of pixels to move the window vertically.

```

<<? <button onclick="openWin()"> Open </button>
<button onclick="moveWin()"> Move </button>
<script>
let myWindow;
function openWin() {
    myWindow = window.open("", "", "width=400, height=400");
}
function moveWin() {
    myWindow.moveBy(250, 250);
}
</script>

```

moveTo()

- ⊛ this method moves a window to the specified coordinates.

```

<<? <button onclick="moveWin()"> Move </button>
<script>
let myWindow;
function moveWin() {
    myWindow.moveTo(500, 100);
}
</script>

```

resizeBy()

- ⊛ this method resizes a window by a specified amount.

- ⊛ Syntax;

resizeBy(width, height)

- * width - Required. A positive or negative number. The number of pixels to resize the width by.
- * height - Required. A positive or negative number. The number of pixels to resize the height by.


```

<button onclick="openWin()">Open </button>
<button onclick="resizeWin()">Resize </button>

```

```

<script>

```

```

let myWindow;

```

```

function openWin() {

```

```

    myWindow = window.open("", "", "width=100, height=100");
}

```

```

function resizeWin() {

```

```

    myWindow.resizeBy(250, 250);
}

```

```

</script>

```

resizeTo()

⊛ this method resizes a window to a new width and height.

⊛ Syntax;

```

window.resizeTo (width, height)

```

```

<button onclick="resizeWin()">Resize </button>

```

```

<script>

```

```

let myWindow;

```

```

function myresizeWin() {

```

```

    myWindow.resizeTo(300, 300);
}

```

```

</script>

```

name

⊛ this property sets or returns the name of the window.

```

<P id="demo"></P>

```

```

<script>

```

```

let name = window.name;

```

```

window.name = "myWindow";

```

```

document.getElementById("demo").innerHTML = name;

```

```

</script>

```

Navigator

- ⊛ this object contains information about the browser.
- ⊛ Syntax;

window.navigator

or

navigator

- ⊛ Navigator object properties;

Property	Description
appName	Returns browser code name
appVersion	Returns browser version
cookieEnabled	Returns 'true' if browser cookies enabled
geolocation	Returns a geolocation object for the user's location
language	Returns browser language
online	Returns 'true' if the browser is online
platform	Returns browser platform
product	Returns browser engine name
userAgent	Returns browser user-agent header

```
<?xml- <p id="demo"></p>
```

```
<script>
```

```
let language = window.navigator.language;
```

```
document.getElementById("demo").innerHTML = language;
```

```
</script>
```

pageXOffset, pageYOffset

- ⊛ pageXOffset property returns the pixels a document has scrolled from the upper left corner of the window.
- ⊛ pageXOffset property is equal to the scrollX property. this is a read-only property.

- ④ pageYOffset property returns the pixels a document has scrolled from the upper left corner of the window.
- ④ pageYOffset property is equal to the scrollTop property.

```

280:- <button onclick = "myfunction()" style = "position: fixed;" > Click Me!
      </button>

      <div style = "background-color: red; height: 2000px; width: 2000px">
      </div>

      <script>
      function myfunction () {
        window.scrollBy (100, 100);
        alert ("X value: " + window.pageXOffset +
              "Y value: " + window.pageYOffset);
      }
      </script>

```

screenX, screenY

- ④ screenX property returns the X coordinate of a window, relative to the screen.
- ④ screenY property returns the Y coordinate of a window, relative to the screen.

print()

- ④ This method prints the content of the current window.
- ④ that means, this method opens a 'Print Dialog Box', which lets the user to select preferred printing options.

```

280:- <button onclick = "window.print()" > Print this page </button>

```

prompt()

- ④ This method displays a dialog box that prompts the user for input.
- ④ This method returns the input value if the user clicks "OK", otherwise returns 'null'.

- ④ Syntax;

prompt (text, defaultText)

- * text - Optional. The text to display in the dialog box.
- * defaultText - Optional. The default input text.


```

<button onclick = "myfunction ()"> Click Me! </button>
<p id = "demo"> </p>
<script>
function myfunction () {
    let person = prompt ("Please enter your name: ", "Harry");
    if (person != null) {
        document.getElementById ("demo").innerHTML =
        "Hello " + person + "! How are you today? ";
    }
}
</script>

```

screen

- ⊛ This object contains information about the visitor's ~~screen~~ screen.
- ⊛ Screen object properties

Property	Description
availHeight	Returns the height of the screen. (excluding the windows taskbar)
availWidth	Returns the width of the screen. (excluding the windows taskbar)
colorDepth	Returns the bit depth of the color palette for displaying images
height	Returns the total height of the screen
pixelDepth	Returns the depth color resolution (in bits per pixel) of the screen
width	Returns the total width of the screen

screenLeft, screenTop

- ⊛ screenLeft property returns the X (horizontal) position of a window,
relative to the screen.
- ⊛ screenTop property returns the Y (vertical) position of a window,
relative to the screen.

```

<?or <p id = "demo"></p>
<script>
let x = window.screenLeft;
let y = window.screenTop;
document.getElementById ("demo").innerHTML =
"Left: " + x + ", Top: " + y ;
</script>

```

separately.

stop()

- ⊛ this method stops window loading, this is same as clicking stop in the browser.

Note: stop() method can be used to stop loading an image, if it takes too long.

eg:-

```
<script>
```

```
  window.stop();
```

```
</script>
```

```
<body>
```

```
<h1> The window object </h1>
```

```
<h2> The stop() method </h2>
```

```
<p> The stop() method stops this document from loading. </p>
```

```
</body>
```